

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and refined **Android SDK applications** for sensor data collection and feature demonstrations, collaborating with a global engineering team to integrate new sensor technologies.
- Engineered cross-platform **desktop tools** for Windows and Linux that interact with Android devices, enabling device control, data validation, and post-processing of collected sensor readings.
- Designed and implemented **Java-based software components** following industry-standard object-oriented design patterns to enhance the functionality of existing sensor data processing systems.
- Contributed to the design and production monitoring of **end-to-end data processing pipelines**, ensuring efficient data flow from device to analysis platforms.
- Supported global internal **data collection campaigns** and customer triage activities by developing robust data acquisition tools and performing root cause analysis on device-related issues.
- Wrote unit and integration tests for **Java components**, achieving over 85% code coverage and ensuring the reliability of core data processing logic.
- Containerized **Java applications** using Docker and managed their deployment on AWS EC2 instances, improving deployment consistency and simplifying environment setup.

SKILLS

Mobile Development – Android SDK, Java, Kotlin, Android Wear / Wear OS, Android XR

Backend Development – Java, Spring Boot, REST APIs, Hibernate, JPA

Programming Languages – Python, C++, SQL

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Azure, GCP

Tools & Technologies – Git, GitHub, Docker, Jenkins, Jira, Confluence, Linux, Splunk, Dynatrace

Data Analytics & BI – Power BI, Tableau, R, pandas, NumPy, scikit-learn, DAX, Power Query

System Design – Object-Oriented Programming, Design Patterns, Microservices

PROJECTS

Android Sensor Data Acquisition Framework

- Developed a core **Android SDK framework** in Java to abstract sensor interactions, enabling rapid development of new data collection applications for diverse hardware.
- Integrated **Bluetooth Low Energy (BLE) communication** for device connectivity and data streaming, handling connection management and data packet parsing for real-time updates.
- Built a companion **Python script** for initial data validation and format conversion, ensuring data integrity before ingestion into downstream processing systems.

Cross-Platform Device Control Tool

- Designed and implemented a **Java desktop application** with a GUI to control and configure Android devices remotely, including capabilities for firmware updates and diagnostic data retrieval.
- Developed a **REST API** interface in Java to manage device connections and command execution, supporting both internal tooling and potential customer integration.
- Utilized **SQL (PostgreSQL)** for storing device configurations and historical diagnostic logs, enabling efficient querying and analysis of device performance.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)